INCH-POUND

MIL-H-44086B
31 March 1992
SUPERSEDING
MIL-H-44086A
28 September 1984

MILITARY SPECIFICATION

HEATER, WATER, LIQUID FUEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable field water heaters capable of supplying heated water, 100°F rise at a rate of 9 gallons per minute. JP-8 fuel or diesel fuel may be used, also alternating current of 208 volts electric power is required (see 6.1).

1.2 Classification. The heater shall be of the following types as specified (see 6.2).

Type I - M-80 Water Heater
Type II - M-85 Water Heater (For use on M-85 Laundry Unit)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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SPECIFICATIONS

FEDERAL

TT-W-572	-	Wood Preservative: Water-Repellent
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
VV-F-800	-	Fuel Oil, Diesel

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-704	-	Treatment and Painting of Materiel
MIL-T-22085	-	Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing
MIL-C-46168	-	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-T-83133	-	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking for U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-6200	-	Heater Assembly, Water, M-80
6-1-6203	-	Section Assembly, Vessel Outside
6-1-6210	-	Firebox Assembly
6-1-9912	-	Heater, Water Assembly, M-85
6-2-2427	-	Duct, Burner Exhaust, Assembly

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(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UXD, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

UNDERWRITERS LABORATORIES, INC. (UL)

Standard No. 372 - Primary Safety Controls for Gas and Oil-Fired Appliances

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

AMERICAN SOCIETY FOR MECHANICAL ENGINEERS (ASME)

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal with the manufacturer's part number identified. The incorporation of such a component in the design of the specified end product shall not require modification to any components, unless such modification is specifically approved by the contracting officer. Prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that the components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Codes and standards. The water heater's flame safeguard control shall comply with the applicable requirements of Underwriters Laboratories (UL) Standard No. 372 (see 4.5).

3.4 Design and construction. The design and construction of the heater shall conform in all respects to the drawings specified in table I, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein. The heater shall operate on 3 phase, 60 hertz, 208 volts, 5 wire electric power source. Three wires are to supply the phases, the white wire is to be 115 volt neutral ground and the green wire is to be the equipment ground. All wiring shall be in accordance with the provisions of NFPA No. 70. Diesel shall be the primary fuel for operation of the heater and shall be in accordance with VV-F-800. JP-8 fuel shall be the secondary fuel and shall be in accordance with MIL-T-83133.

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TABLE I. Lead drawings

Type I, M-80 Water Heater	Drawing 6-1-6200
Type II, M-85 Water Heater	Drawing 6-1-9912

3.5 Performance.

3.5.1 Firebox assembly. Each firebox assembly (Drawing 6-1-6210) shall have no visible evidence of leakage when tested as specified in 4.4.5.

3.5.2 Outside vessel section. Each outside vessel section (Drawing 6-1-6203) shall have no visible evidence of leakage when tested as specified in 4.4.5.

3.5.3 Heater assembly. The heater assembly shall be put in operation without the use of tools, except for a 7/8 inch wrench to attach the fuel lines to the drum fill assembly. The heater assembly shall operate properly without any indication of burner pulsation, flame failure or visible smoke. Testing shall be as specified in 4.4.5.

3.5.4 Fuel pump.

3.5.4.1 Fuel pump, type I (M-80). The fuel pump shall be adjusted to operate at a pressure of 100 ± 2 psig. Testing shall be as specified in 4.4.5.

3.5.4.2 Fuel pump, type II (M-85). The fuel pump shall be adjusted to operate at a pressure of 125 ± 2 psig for elevations up to 2,500 feet above sea level, 115 ± 2 psig for elevations over 2,500 to 4,000 feet above sea level, or 100 ± 2 psig for elevations over 4,000 feet above sea level with a 4 GPH nozzle. Testing shall be as specified in 4.4.5.

3.5.5 Burner smoke. There shall be no visible smoke at the pipe stack outlet (or exhaust duct) one minute after ignition of the burner, when tested as specified in 4.4.5.

3.5.6 Water level control. The water level control shall prevent operation of the burner and sound the buzzer whenever the water level in the heater is below the end of the water level electrode, when tested as specified in 4.4.5. Also the burner shall restart and the buzzer shall stop when the water level is raised to the operating condition, when tested as specified in 4.4.5.

3.5.7 Operating control. The operating control shall be located on the probe box and wired in series with the limit control. The control shall be provided with a calibrated knob and dial in degrees Fahrenheit. The control

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shall be adjustable over a temperature range of 0° to 250°F and shall be adjusted and set per the manufacturers instructions prior to performance testing. The control shall shut down the burner when the internal water temperature reaches a set temperature (+0°F, -5°F) and the burner shall start when the internal water temperature falls below a set temperature (+5°F, -0°F) when tested as specified in 4.4.5.

3.5.8 Limit control. The limit control shall be located in the upper manifold and wired in series with the operating control. The control shall be factory preset at 190°F and shall be normally closed and shall open on a temperature rise above the preset temperature. The control shall halt the burner cycle when the internal water temperature rises above 190°F (+10°F, -0°F) and initiate the burner cycle when the internal water temperature falls below 190°F (+0°F, -10°F), when tested as specified in 4.6.8.

3.5.9 Flame safeguard. The flame safeguard shall operate between 125°F and -20°F. The scanner component shall operate between 212°F and -40°F. Prior to each start the control shall provide a 7-second purge cycle. Following each purge cycle, a 10-second trial for ignition shall occur. When flame failure occurs the control shall de-energize the fuel solenoid, shut off the burner motor, provide a mechanical lockout and turn on the alarm circuit. For type I (M-80), when flame failure occurs the control shall also shut off the ignition transformer. The burner motor shall not shut off until a pre-set time delay is completed for post-purge of the burner/heat exchanger assembly.

3.5.9.1 Flame failure, initial start lockout. Flame failure initial start lockout shall observe the following sequence:

a. A purge cycle of 7 (± 2) seconds.
b. A trial of 10 (± 2) seconds for ignition when the solenoid valve is opened and then closed.
c. A lockout shall occur in not greater than 70 seconds from the time the water heater switch is turned on.
d. A post purge of 120 (± 10) seconds.

When tested as specified in 4.4.5, lockout time shall be not greater than 70 seconds.

3.5.9.2 Firing period lockout. The firing period lockout requirements shall observe the following sequence upon flame failure:

a. A 4-second (maximum) flame failure response, when the solenoid valve closes.
b. A purge cycle of 10 (± 2) seconds.
c. A trial of 10 (± 2) seconds for ignition when the solenoid is opened and then closed.

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d. Not greater than a 70-second mechanical lockout occurrence from the time the flame was extinguished
e. A post purge of 120 (± 10) seconds.

When tested as specified in 4.4.5, lockout time shall be not greater than 70 seconds.

3.5.9.3 Flame signal. The flame control shall not be effected by the ignition spark and shall have a flame failure response of less than 4 seconds when tested as specified in 4.6.11.

3.5.10 Signal device. The signal device shall sound to indicate low water or flame safe guard lockout. Testing shall be as specified in 4.4.5.

3.5.11 Circuit breaker. All the circuit breaker shall operate without damage to themselves or other electrical components.

3.5.11.1 Line starter. The line starter shall open within a one minute period at a current flow of 8 amps and shall be reset by means of a reset button when tested as specified in 4.6.13.

3.5.11.2 Burner motor. The burner motor shall be protected by a thermal overload protector that is an integral part of the motor. The motor shall shut off in 25 seconds and be capable of resetting without the use of tools when tested as specified in 4.6.14.

3.5.12 Relief valve. The relief valve shall open at a pressure of 50 to 63 psig. After the valve has opened and pressure decreases, the valve shall close before a pressure of 25 psig is reached when tested as specified in 4.6.15.

3.5.13 Heater capacity. The heater shall provide a flow of 9 gallons per minute with a temperature rise of not less than 100°F when tested as specified in 4.4.5.

3.5.14 Water heater assembly. Each water heater assembly (type I - Drawing 6-1-6200 and type II Drawing 6-1-9912) shall have no visible evidence of leakage or improper operation when tested as specified in 4.4.5.

3.6 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Interchangeability. All component parts of any one heater shall be so fabricated as to be interchangeable with the like parts of any other unit furnished under any one contract.

3.8 Workmanship. The heater components shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malformations. There shall be no sharp edges, burrs or slivers. The finish

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shall be free of blisters, peel, chipped areas, or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray sag, imprint, or embedded foreign matter. All bolts, studs, nuts, and screws shall be drawn tight. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

3.8.1 Welders. Unless otherwise specified (see 6.2), before assigning any welder to work, the contractor shall provide the contracting officer, or his authorized representative, with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by either of the following listed codes for the type welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

a. Standard Qualification Procedure of the American Welding Society.
b. Welding Qualification of the American Society of Mechanical Engineers.

3.8.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds and rough edges and projections shall be dressed smooth.

3.8.3 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack, thereby providing no strain on wires or terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested in sequential order as specified in 4.4.5. The first article shall be tested in accordance with 4.6.3 for 60 hours of continuous operation.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the UL Flame Safeguard Control, NFPA wiring, and welders conforms to the requirements specified in 3.3.1, 3.4, and 3.8.1.

4.4.2 In-process inspection. In-process examination shall be made of surface preparation prior to application of the finished coat to determine compliance with applicable finish requirements. Also examination of the surface preparation prior to welding shall be made to determine compliance with 3.8.2. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

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4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. Visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified		201
	Blistered, peeled, chipped, or area of no film		202
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded		203
Design, construction, and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	101	
	Fractured, split, punctured, torn, dented, creased deteriorated, or malformed	102	
	Component misplaced or not in proper alignment		204
	Component loose		205
	Sharp burr or edge or sliver		206
	Any operation omitted or not properly performed	103	
	Any functioning component that is inoperative or will not operate as intended	104	
	Any functioning component that requires abnormal force to operate or not interchangeable		207
	Any adjustable assembly that cannot be adjusted		208
	Any assembly not properly adjusted		209
	Any component not readily accessible for servicing where required		210
Welding	Missing, incomplete, cracked, fractured, not fused	105	

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TABLE II. Visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Welding (cont'd)	Undercut, not smooth and uniform, or scale or flux deposits not removed		211
Soldering (when applicable)	Missing, not adherent, or incomplete		212
	Not clean (flux residue not removed, pinholes in solder)		213
Bolts, nuts, screws, studs, and other types of threaded fasteners	Broken, stripped, fractured, or not tight		214
Rivets (when required)	Broken, not peened or insufficiently peened, or loose		215
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	106	
	Adequate slack not provided for wiring to relieve strain		216
Gauges, instruments, controls, thermometer, etc.	Not properly mounted or not specified type or capacity		217
	Not free of foreign matter		218
Marking	Missing, incomplete, not legible, not specified type or size, not in proper location	107	

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of heaters. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. Each production end item shall be tested for the characteristics listed in table III in the order as listed. Failure of any test shall be cause for rejection of the end item involved. All heater function tests, 4.6.4, 4.6.5, 4.6.6, 4.6.7, 4.6.9, 4.6.10, 4.6.12, and 4.6.16

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may be performed using test supply line, test return line, and test stack outlet pipe for type I water heaters or test exhaust duct for type II water heaters in lieu of equipment that will be furnished with each unit. Each production unit tested per 4.6.3 shall be tested for 4 hours of continuous operation.

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Firebox assembly (leakage)	3.5.1	4.6.1
Outside vessel section (leakage)	3.5.2	4.6.2
Heater assembly	3.5.3	4.6.3
Fuel pump (pressure)	3.5.4	4.6.4
Burner smoke	3.5.5	4.6.5
Water level control	3.5.6	4.6.6
Operating control	3.5.7	4.6.7
Flame failure, initial start lockout	3.5.9.1	4.6.9
Firing period lockout	3.5.9.2	4.6.10
Signal device	3.5.10	4.6.12
Heater capacity	3.5.13	4.6.16
Water heater assembly (leakage)	3.5.14	4.6.17

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	In adequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.5 UL Standard compliance. Acceptable evidence of meeting the applicable requirements of the Underwriters Laboratories Standard specified in 3.3.1 shall be a UL listing mark, or a certified test report from a recognized independent testing laboratory indicating that the heater's flame safeguard control has been tested as specified and conforms to UL Standard No. 372. Such evidence must be acceptable to the contracting officer or his authorized representative prior to the start of production.

4.6 Methods of inspection. Commercial grade No. 1 or No. 2 diesel fuel or JP-8 fuel can be used as the fuel for the applicable tests.

4.6.1 Firebox assembly (leakage) test. Each firebox assembly shall be separately tested for leaks by applying an internal compressed air pressure of 75 psi for a period of 5 minutes. The unit shall be submersed in water for the 5 minute period to detect the location of any leaks. Evidence of leakage shall be any decrease in air pressure within a 5 minute period or the appearance of any air bubbles excluding surface trapped air. Any nonconformance with the requirements of 3.5.1 shall constitute failure of this test.

4.6.2 Outside vessel section (leakage) test. Each outside vessel section shall be separately tested for leaks by applying an internal compressed air pressure of 75 psi for a period of 5 minutes. The unit shall be submersed in water for the 5 minute period to detect the location of any leaks. Evidence of leakage shall be any decrease in air pressure within the 5 minute period or the appearance of any air bubbles excluding surface trapped air. Any nonconformance with the requirements of 3.5.2 shall constitute failure of this test.

4.6.3 Heater assembly test.

4.6.3.1 Type I only. One water heater assembly shall be placed in operation. Any nonconformance with the assembly and operation requirements of 3.5.3 shall constitute failure of this test.

4.6.3.2 Type II only. One water heater assembly shall be set up and in a position 3-1/2 feet from the ground. Two exhaust ducts (in accordance with drawing 6-2-2427) shall be attached to the water heater and the other end shall rest on the ground as to exactly duplicate installation on the M85 Laundry Unit. The test shall be conducted outdoors (i.e. without the aid of an exhaust duct fans or shop exhaust systems). Water at 33°F to 50°F shall be provided to the water heater by the water pump. The operating control shall be set up at 100°F greater than the temperature of the water supplied. A system (automatic or manual) shall be set up to draw 30 gallons (± 1 gallon) of hot water from each water heater every 10 minutes for the duration of the test. Once the test is initiated NO adjustments shall be permitted. Any nonconformance with the requirements of 3.5.3 shall constitute failure of this test.

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4.6.4 Pump fuel (pressure) test. The fuel pump gage shall read a pressure in accordance with 3.5.4 anytime during burner operation. Any nonconformance with the requirements of 3.5.4 shall constitute failure of this test.

4.6.5 Burner, smoke test. The heater burner shall be operated and checked for visible smoke. Any nonconformance with the requirements of 3.5.5 shall constitute failure of this test.

4.6.6 Water level control test. Operate the water heater for at least one minute. Shut off pump and drain water heater through the water inlet until signal device sounds an alert Verify that the water level is just below the level control electrode. Close the drain and start the pump. Any nonconformance with the requirements of 3.5.6 shall constitute failure of the test.

4.6.7 Operating control test. The operating control shall be tested with the heater in operating condition. The operating control shall be set at 140°F and the limit control by-passed. The burner shall be operated in accordance with 3.5.4 with the specified 5 GPH nozzle and adjusted below the smoke point. A constant water flow of 5 ± 0.5 gallons per minute thru the heater shall be maintained. After equilibrium has been reached the unit shall be allowed to operate thru three continuous cycles. Repeat the test with the operating control set at 180°F. The thermo-switch on the operating control may be adjusted to meet the water outlet temperature requirements in 3.5.7. Any nonconformance with the requirements as specified in 3.5.7 shall constitute failure of this test.

4.6.8 Limit control test. The limit control shall be tested with the heater in operating condition. The operating control shall be by-passed and a constant flow of 5 GPM ± .5 maintained thru the heater. The burner shall be operated in accordance with 3.5.4 with the specified 5 GPH nozzle. The burner shall be adjusted slightly below the smoke point. After equilibrium has been reached the unit shall be operated thru three continuous cycles. Any nonconformance with the requirements in 3.5.8 shall constitute failure of this test.

4.6.9 Initial start lockout test. During initial starting of the water heater the fuel valve shall be kept closed and the unit tested for initial flame lockout. Any nonconformance with the start sequence or the lockout time of not more than 70 seconds as specified in 3.5.9.1 shall constitute failure of this test.

4.6.10 Firing period lockout test. The burner shall be operated for more than 2 minutes. The fuel valve shall be closed and the unit shall be tested for flame failure during the firing period. Any nonconformance with the flame failure response sequence or the lockout time of not more than 70 seconds as specified in 3.5.9.2 shall constitute failure of this test.

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4.6.11 Flame signal test. An AC/DC multimeter shall be connected between terminals 2 and 3 of the Flame Safeguard Control (UVM-2). When flame is detected, a nominal reading of 115 volts AC should be obtained. When the flame is extinguished, and the spark is on, a reading of 0 volts should be obtained within 4 seconds. Any nonconformance with the control requirements of 3.5.9.3 shall constitute failure of this test.

4.6.12 Signal device test. The buzzer shall sound to indicate low water in 4.6.6 and flame safe guard lockout in 4.6.9 and 4.6.10. Any nonconformance with the requirements of 3.5.10 shall constitute failure of this test.

4.6.13 Line starter test. A current of 8 amps shall be applied to the load side of the circuit breaker. The circuit breaker shall open within a one minute period and after 5 minutes of cooling the breaker shall be reset by closing the switch. Any nonconformance with the requirements of 3.5.11.1 shall constitute failure of this test.

4.6.14 Burner, motor test. A locked rotor test shall be performed to demonstrate that the burner motor thermal overload will activate within 25 seconds of a locked rotor application. After 5 minutes of cooling the overload switch shall be reset by depressing the button. Any nonconformance to the requirements as specified in 3.5.11.2 shall constitute failure of this test.

4.6.15 Relief valve test. The relief valve shall be tested for opening and closing while installed on the water heater. Attach a high pressure water pump with a pressure regulator and gage to the water heater. Fill with water then close all outlet valves on the water heater. Slowly raise the water pressure until the relief valve opens then slowly decrease the water pressure until the relief valve closes. Repeat test three times in succession. Any nonconformance with the requirements of 3.5.12 shall constitute failure of this test.

4.6.16 Heater capacity test. With the operating control by-passed, and a constant flow of 9 gallons per minute thru the heater, the inlet and outlet temperatures shall be recorded. Temperature rise exceeding 100°F shall be maintained once the maximum temperature has been reached. Any nonconformance with the requirements of 3.5.13 shall constitute failure of this test.

4.6.17 Water heater assembly (leakage) test. Each water heater assembly shall be separately tested for leaks under a hydrostatic pressure of 75 psi gage for a period of 5 minutes. To examine the firebox for leakage the front cover and burner assembly shall be removed. Any nonconformance with the requirements of 3.5.14 shall constitute failure of this test. After successfully satisfying the requirements of 3.5.14, the front cover and burner assembly shall be replaced and the heater capacity (see 4.6.16) test shall be repeated. Any improper operation or any evidence of an improper seal around the front cover or any nonconformance with 3.5.13 shall constitute failure of this test.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Drainage of water system. The entire water heater shall be thoroughly drained, blown out and dried. All valves and draincocks shall be left open.

5.1.1.2 Fuel system. The fuel supply and return lines shall be connected to the oil filter and fuel pump, respectively. The hose shall be coiled beneath the heater and the ends connected to the fuel line holder. The burner pump shall be filled with P-10, grade 10 or grade 30 preservative of MIL-P-116.

5.1.1.3 Cleaning, drying, and preservative application. All uncoated and unpainted surfaces of the heater shall be cleaned in accordance with MIL-P-116, process C-1, and thoroughly dried prior to application of preservative. Preservative specified in 5.1.1.5 shall conform to the applicable specifications listed in MIL-P-116. Barrier material specified in 5.1.1.5 shall conform to type I or II, grade A, class 2 of MIL-B-121. Tape specified hereinafter shall conform to type II of MIL-T-22085.

5.1.1.4 Repainting and recoating. Painted and permanently coated surfaces which are defective or from which the paint or permanent coating has been removed, shall be repainted or recoated with material identical with the original.

5.1.1.5 Heater assembly. The heater assembly shall be drained and dried of water and fuel as specified in 5.1.1.1. All interior surfaces that are unlined, or not treated to resist corrosion, shall be coated with P-10, type I, grade 30 preservative of MIL-P-116 applied by fogging, spraying, or completely filling and draining. Excess preservative shall be drained and the drain and fill plugs, before being reinstalled, shall be coated with the preservative. All water line openings on the heater assembly shall be sealed with tape. All openings on the air shutter, smoke stack and electric motor shall be sealed with barrier material and tape, or by tape only, as applicable. All gages, dials, and sight glasses not protected by a cover, shall be cushioned, overwrapped with barrier material, and the wrap secured in place with tape.

5.1.1.6 Drum fill adapter assembly. The drum adapter extension shall be threaded, hand-tight into the 1/4 inch return port.

5.1.2 Commercial preservation. The heater shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Heater assembly and lower smoke stack assembly. The heater, and drum fill adapter, preserved as specified in 5.1, shall be packed in snug-fitting nailed wood box constructed in accordance with class 2, grade A, style 7, type 3 load of PPP-B-621. The water heater shall be bolted to the box bottom through the skids. Additional skids are required and shall be located on the box bottom adjacent to the inner edges of the outside skids to allow for bolting of the water heater through the skids. For type I (M-80) only, the elbow and lower stove pipe section of the smoke stack assembly shall be attached to the boiler. The fuel lines shall be coiled beneath the heater and the ends attached to the fuel line holder. Assembly of the box shall be in accordance with the appendix of PPP-B-621, except that the sheeting on the ends shall be attached to the interior frame members with nails and to the side of the skids with screws. All wood surfaces shall be cleaned and treated in accordance with type B of MIL-T-704, using TT-W-572, composition B. The exposed metal surfaces shall be cleaned and treated in accordance with type F or type G of MIL-T-704. All exterior surfaces shall be painted camouflage green 383 conforming to type II of MIL-C-46168. The word "CARC" and the month and year of application shall be stencilled in 1 inch block letters, using lusterless black paint conforming to type II of MIL-C-46168, on the outside of the container directly below the marking required in 5.3.1. The assembled and painted box shall be strapped as specified in the appendix of PPP-B-621.

5.2.2 Commercial packing. The heater preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Additional marking. For level A shipping containers as specified in 5.2.1.1, the following additional marking shall be stencilled on top of the container in 1 inch high block letters using lusterless white paint conforming to type II of MIL-C-46168.

REUSABLE CONTAINER

DO NOT DESTROY

TO OPEN REMOVE SCREWS

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The water heater is intended to provide hot water for the military field sanitation, laundry and bath units.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2 and 3.4).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. When certification of qualification test for welders is not required (see 3.8.1).
f. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Field
Hot water
M-80
M-85
Portable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CS

Preparing activity:

Army - GL

(Project 4520-0330)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-44086B	1992 March 31

3. DOCUMENT TITLE

HEATER, WATER, LIQUID FUEL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4531	256-4531
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-UXT Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Previous editions are obsolete.

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